

Hadi Lashkari Ghouchani

Software Engineer • ML Systems

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Summary

Software engineer with 21+ years of development experience and an M.Sc. in Theoretical Physics. Independently designed, implemented, and benchmarked a new neural-network architecture (type-nn) in C and PyTorch, with custom autograd, a per-unit optimizer, structure learning, and a GPU backend. Brings 11 years of production engineering, most recently CI/CD infrastructure, build reproducibility, and multi-agent LLM workflows.

- Takes ML ideas from the math to a C reference, a PyTorch library, and a GPU backend, verified number for number
- Brings engineering rigor to research: reproducible environments, pinned datasets, deterministic multi-seed benchmarks
- Comfortable deriving and debugging gradients, numerical stability, and optimization dynamics
- Built production multi-agent LLM workflows (planner, coder, reviewer) in CI
- Taught algorithms and programming at university level
- Communicates technical decisions openly, through RFCs, write-ups, and blog posts

Looking to move into empirical AI research, focused on ML systems and on making AI systems reliable and beneficial.

Machine Learning Research & Engineering

Type Neural Network: [type-nn \(C\)](#) and [torch-type-nn \(PyTorch\)](#)

Independent research  2025 - Present  Toronto, Canada

A neural network whose layers are partition functions of an And over Ors (product and sum types), derived from my [Type Mechanics](#) framework. The network grows and prunes its own width, degree, and depth while it trains, so there is no width or depth hyperparameter to choose. Write-up: [Train the knowledge](#).

- Wrote the C reference implementation and a PyTorch port published on PyPI; forward pass, every gradient, and optimizer steps agree with C to 1e-12, enforced by a test that compiles and runs the C code
- Implemented a custom autograd Function that computes exact gradients in log space, including one-sided derivatives at exact zeros where standard autograd returns NaN
- Built a per-unit Adam optimizer whose moments move with the weights through structural edits, and made stock torch.optim optimizers work by remapping their state through index maps
- Designed an architecture-agnostic structure scaler (protocol + adapters) that also grows and prunes standard MLPs, with a threshold rule and a Bayesian-information-criterion rule
- Batched variable-size layers with exact identity padding for GPU training, and packed ragged layers into a fused GEMM; the GPU test suite fails if it silently falls back to CPU
- Ported the C benchmark board exactly (same splits, shuffles, learning rates, 5 seeds per cell) and reproduced it digit for digit; the whole pipeline runs from one Nix flake

Matrix Fast Fourier Transform (MFFT)

Independent research  2024 - Present  Toronto, Canada

Proposed an FFT-based optimization for matrix multiplication ([post](#)) and built [mfft-bench](#), a benchmark comparing matmul implementations.

Applied ML competitions

- 2024 First place, National Health Datathon
- 2024 Second place, TMLS Hackathon: a [retrieval-augmented LLM assistant](#) over City of Toronto data

Experience

Senior Mobile Developer

at [Bell Canada \(via Hays\)](#)  April, 2026 - Present  Toronto, Canada

Bell Canada is Canada's largest telecommunications company. I own the CI/CD infrastructure for its Android and iOS apps.

Mission:

- Built a multi-agent LLM workflow in CI that turns Jira tickets into merge requests, with planner, coder, and reviewer agents committing an initial implementation
- Took ownership of GitLab CI/CD and reversed a declining trend in pipeline job success rate, raising it from 18% to 31% (a 72% relative improvement) within six months
- Made builds and emulator runs reproducible by wrapping them in Nix flakes, and migrated jobs to custom, task-specific Linux containers for performance and isolation
- Designed trunk-based development pipelines for Android and iOS across multiple repositories, automating release branches, cherry-picks, publishing, and release-candidate promotion
- Restructured merge request pipelines into fast checks, required heavy jobs, and optional jobs, shortening feedback loops
- Automated DTO and mapper generation from API specs in CI, removing hand-written boilerplate
- Reduced GitLab SAST security findings to zero across multiple projects
- Maintain and tune the AWS EC2 instances running CI/CD for reliability and performance

Technologies: Android, Kotlin Multiplatform, iOS, Swift, GitLab CI, GitLab SAST, Podman, Nix, AWS EC2, LLM agents

Senior Mobile Developer

at [Block Inc. - Bitkey project \(via Touchlab\)](#)  November, 2025 - Feb, 2026  Toronto, Canada

Bitkey is Block Inc.'s self-custody bitcoin wallet, available in the U.S. and 95 other countries.

Mission:

- Fixed flaky unit and functional tests by diagnosing and resolving multiple race conditions and state management problems
- Worked on the multi-backup project and the new onboarding flow

Technologies: Kotlin Multiplatform, Compose Multiplatform, Swift, Rust bindings (UniFFI), Ktor, SqlDelight, Kotest, BuildKite

Lead Android Developer

at [CIBC \(via Vision Talent\)](#)  March, 2025 - November, 2025  Toronto, Canada

The Canadian Imperial Bank of Commerce is one of Canada's largest banks.

Mission:

- Cut the shared CI pipeline build time by almost 50%, mainly through build caching
- Initiated and implemented memory management by applying structural DI scopes to release resources, ViewModels, and their state
- Rewrote features in the new CIBC mobile apps and worked on remote feature flags

Technologies: Kotlin Multiplatform, Compose Multiplatform, SwiftUI, Ktor, Koin, Jenkins

Software Engineer -> Productivity Engineer

at [Bell Canada \(via MobileLIVE\)](#)  Apr. 2021 - Nov. 2024  Toronto, Canada

Promoted to Productivity Engineer, supporting a team of 50+ Android developers.

Mission:

- Resolved VPN proxy and self-signed certificate issues with an automated installer and a Compose Multiplatform app, increasing builds on the Develocity dashboard by 164%
- Reduced incremental test runtime by 52%
- Introduced Gradle convention plugins and a single source of truth for library versions, significantly reducing crashes from version conflicts
- Introduced an RFC process for technical decisions and led regular knowledge-sharing sessions on SOLID, modularization, feature flags, and monorepos
- Spearheaded Merge Trains and Merge Request Pipelines in GitLab, and integrated Detekt code-quality checks

Staff Android Developer

at [iFood \(via Letgo\)](#)  Oct. 2020 - Mar. 2021  Barcelona, Spain

Latin America's leading food delivery platform (50M+ Android installs). Refactored legacy code out of the root module, cutting incremental build time by 55%.

Senior Android Developer

at [Letgo](#)  Oct. 2019 - Mar. 2021  Barcelona, Spain

One of the largest classified marketplaces in the U.S. at the time (15M+ Android installs in Turkey). Optimized feed loading, reducing launch time by over 60 seconds.

Lead Android Developer

at [Sheypoor](#)  Jan. 2018 - Jan. 2019  Tehran, Iran

Classified marketplace with 13M+ users. Led initiatives that tripled daily active users from 500K to 1.5M+ within one year.

Senior Android Developer

at [Myket: App Store and Social Network](#)  Sep. 2015 - Dec. 2017  Tehran, Iran

Java and C# developer

at [Soha System](#)  Mar. 2015 - Aug. 2015  Tehran, Iran

Other Public Contributions

-  2026 Tabular center: State machines whose declaration is the transition matrix
-  2025 Post: Geometrical Probability
-  2024 C Numbers: Number types in C, including floating-point and p-adic numbers
-  2024 Dragon detector: An image classification model deployed on Replicate
-  2024 Nix Effect pod: Isolating side effects with NixOS and podman
-  2024 Post: Not all code is a liability
-  2023 Contributor to nixpkgs/androidenv

Education

Sharif University Of Technology	M.Sc	in Physics	 Feb. 2010	 Tehran, Iran
Payame Noor University	B.Sc	in Physics	 Jul. 2007	 Mashhad, Iran

Tools / Skills

Machine Learning

PyTorch: custom autograd Functions, nn.Module, custom optimizers, state_dict migration
GPU execution with PyTorch/CUDA, fused GEMM
Structure learning: growing/pruning networks
Model selection (BIC), multi-seed evaluation
Numerical stability, log-space gradients
LLM APIs, RAG, multi-agent workflows

Research Engineering

Reproducible experiments: Nix flakes, pinned data
Deterministic benchmarks and result boards
Cross-implementation numerical verification
pytest, ruff, PyPI packaging (sdist/wheel)

Mathematics / Physics

M.Sc. Physics, Sharif University of Technology
Linear algebra, probability, FFT
Statistical mechanics / partition functions

Programming Languages

Python/C/Rust/Kotlin/Java/Scala/Swift
Groovy/Dart/JavaScript/Ruby/Bash

Infrastructure / CI

Linux (NixOS/Fedora/Ubuntu/Debian), MacOS
Nix/NixOS, Podman/Docker, Buildkit
AWS EC2, GitLab CI, GitHub Actions
BuildKite, Jenkins, Develocity
GitLab SAST, dependency pinning, caching

Systems Programming (Rust)

tokio/crossbeam/Rc/Arc/Mutex
FFI/UniFFI, tonic/prost, flatbuffers/thrift

Software Design

Clean/Hexagonal/Unidirectional Architecture
SOLID, Functional Programming, Stateless design
Modularization, Stable Dependencies/Abstractions

Mobile (11 years)

Android, Kotlin Multiplatform, Compose
iOS, Swift/SwiftUI
Gradle/Kotlin compiler/KSP/Detekt plugins
Main Contributor of Log4K, LifecycleHandler

Workflow

Trunk-based development, RFC process
Cross-functional teams, mentoring

Achievements and Recognition

- 2024 Second place, AWS Summit GameDay
- 2018 Result-oriented badge, Sheypoor
- 2007 Ranked 44th of 5,000+ participants in the National Graduate Entrance Exam in Physics

Language

English: Advanced
Farsi: Native

The latest version of this resume is available in [github](#)
Built on September 29, 2026